Top500 Dataset — General Statistics and Geometric Analysis

Residues: 97,519 | **Chains:** 514 | **Structures:** 492 | **Source:** top500_measures.jsonl | **Generated by:** pydangle-biopython v0.5.1

This document provides the following analyses:

1. **General Information** — dataset overview, measurements available, and summary statistics
2. **Amino Acid Composition and Chain Length Distribution** — comparison with UniProt and PDB reference databases
3. **Torsion Angle Distributions** — univariate (backbone, omega deviation, sidechain) and multivariate (Ramachandran) distributions using circular statistics
4. **Bond Angle Distributions** — tau (N−Cα−C) and other non-periodic angles using linear statistics
5. **References**

1. General Information

Dataset	Top500
Total residues	97,519
Unique structures	492
Unique chains	514

Measurements available: phi: 94,594, psi: 94,594, omega: 94,594, tau: 97,519, chi1: 81,072, chi2: 60,606, chi3: 36,711, chi4: 17,327. Not all residues have all measurements — terminal residues lack phi/psi/omega, and glycine and alanine lack sidechain chi angles.

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Ramachandran category distribution:
  General        65,472  (67.14%)
  IleVal         12,041  (12.35%)
  Gly             7,275  ( 7.46%)
  TransPro        3,927  ( 4.03%)
  PrePro          3,143  ( 3.22%)
  CisPro            219  ( 0.22%)

Torsion angles (circular statistics):
  Angle             N    Circ Mean     Circ Std
  -------- ---------- ------------ ------------
  phi          94,594      -85.589       42.247
  psi          94,594       13.349      115.731
  omega        94,594      179.505        7.300
  chi1         81,072      -87.483       85.503
  chi2         60,606      172.912      117.293
  chi3         36,711     -172.964      103.976
  chi4         17,327       -3.292      103.523

Bond angles (linear statistics):
  tau          97,519      111.000        3.190
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2. Amino Acid Composition and Chain Length Distribution

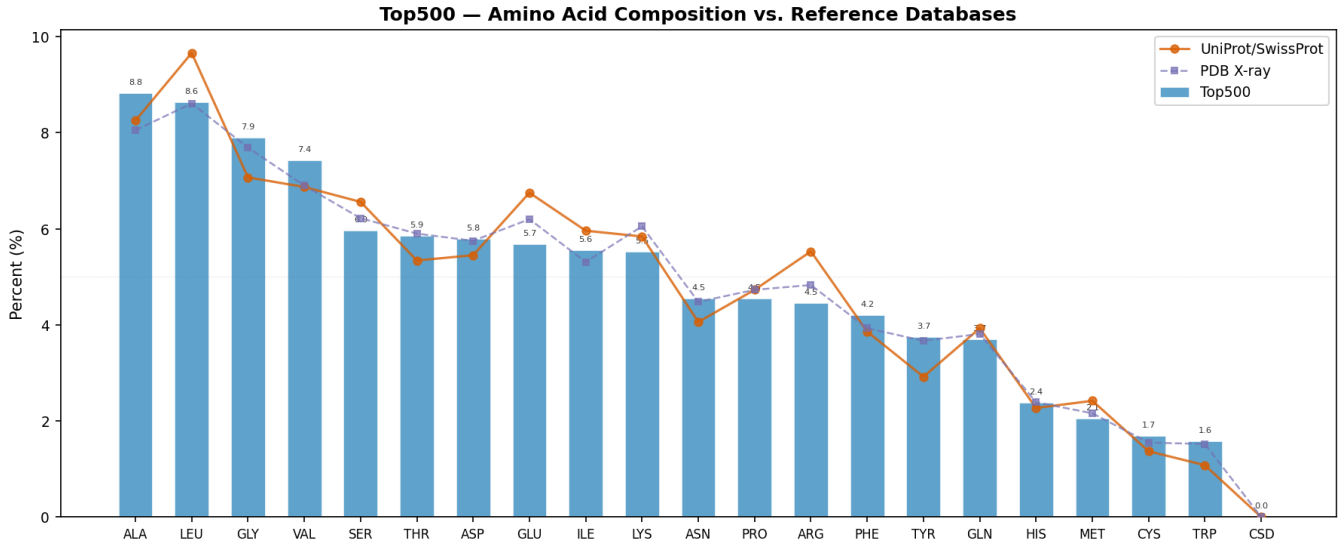
2.1 Representativeness

The figure and table below compare the Top500 amino acid frequencies against two reference databases:

- **UniProt/SwissProt** (~2024): amino acid frequencies across all reviewed protein sequences, representing the broadest available view of protein sequence space.
- **PDB X-ray** (~2025): frequencies from a 10,000-entity sample of X-ray crystallographic structures in the RCSB Protein Data Bank.

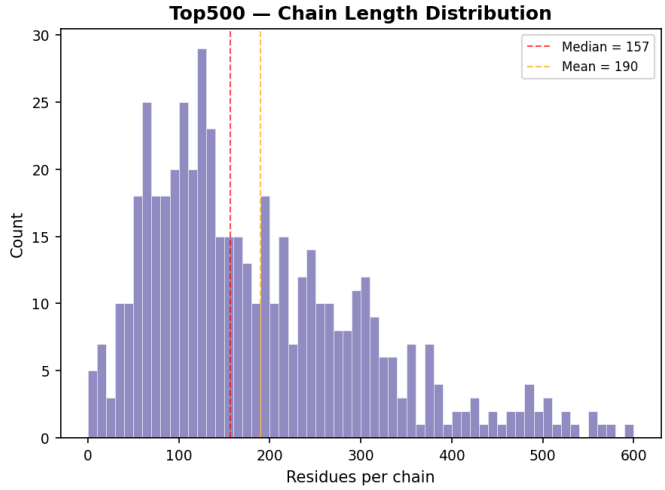
The two reference distributions are highly similar (most amino acids differ by <0.5%), indicating that the PDB's well-known crystallization bias — overrepresentation of soluble, globular, well-expressing proteins from model organisms — has only a modest effect on overall amino acid composition.

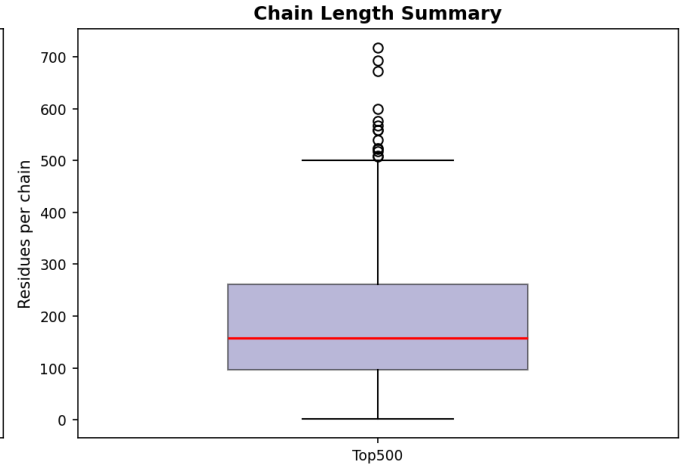
The Top500 dataset (97,519 residues from 492 structures) shows additional deviations from both references due to quality filtering, homology reduction, and sample size effects.



Residue	Count	Dataset%	UniProt%	PDB%	Δ UniProt
ALA	8,600	8.82%	8.25%	8.05%	+0.57%
LEU	8,421	8.64%	9.66%	8.61%	-1.02%
GLY	7,704	7.90%	7.07%	7.69%	+0.83%
VAL	7,243	7.43%	6.87%	6.90%	+0.56%
SER	5,811	5.96%	6.56%	6.22%	-0.60%
THR	5,711	5.86%	5.34%	5.90%	+0.52%
ASP	5,643	5.79%	5.45%	5.75%	+0.34%
GLU	5,533	5.67%	6.75%	6.20%	-1.08%
ILE	5,415	5.55%	5.96%	5.31%	-0.41%
LYS	5,380	5.52%	5.84%	6.05%	-0.32%
ASN	4,431	4.54%	4.06%	4.48%	+0.48%
PRO	4,429	4.54%	4.73%	4.73%	-0.19%
ARG	4,349	4.46%	5.53%	4.83%	-1.07%
PHE	4,096	4.20%	3.86%	3.93%	+0.34%
TYR	3,654	3.75%	2.92%	3.67%	+0.83%
GLN	3,601	3.69%	3.93%	3.81%	-0.24%
HIS	2,316	2.37%	2.27%	2.39%	+0.10%
MET	2,004	2.05%	2.42%	2.16%	-0.37%
CYS	1,645	1.69%	1.37%	1.55%	+0.32%
TRP	1,532	1.57%	1.08%	1.52%	+0.49%
CSD	1	0.00%	0.00%	0.00%	+0.00%

2.2 Chain Length Distribution



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Chain length statistics (514 chains):
  Min:              1
  Q1:              96
  Median:         157
  Mean:         189.7
  Q3:             260
  Max:            718
  Std:          126.4
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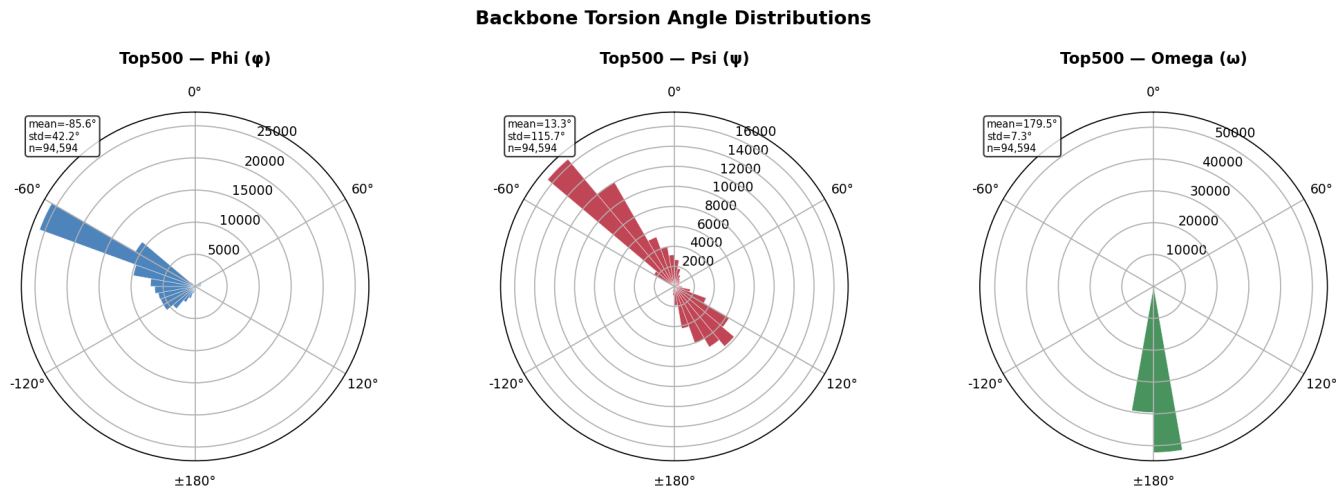
3. Torsion Angle Distributions (Circular Statistics)

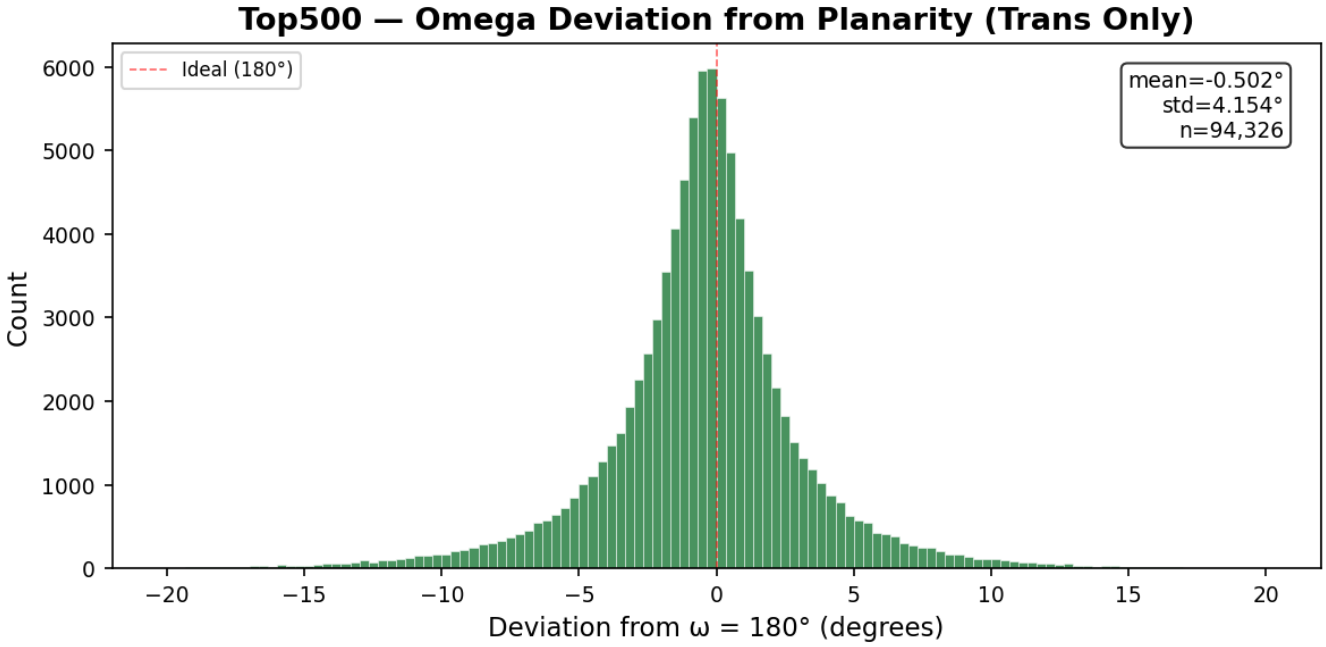
Torsion angles (φ, ψ, ω, χ) are periodic variables defined on the interval [−180°, +180°]. Standard (linear) arithmetic means and standard deviations give misleading results when values cluster near the ±180° boundary — for example, a linear mean of trans peptide bond ω values near +179° and −179° yields ~0° rather than the correct ~180°. All torsion angle summary statistics in this report use **circular statistics**: the circular mean is computed as atan2(⟨sin θ⟩, ⟨cos θ⟩), and the circular standard deviation as $\sqrt{(-2 \ln \bar{R})}$ where $\bar{R}$ is the mean resultant length.

3.1 Univariate Distributions

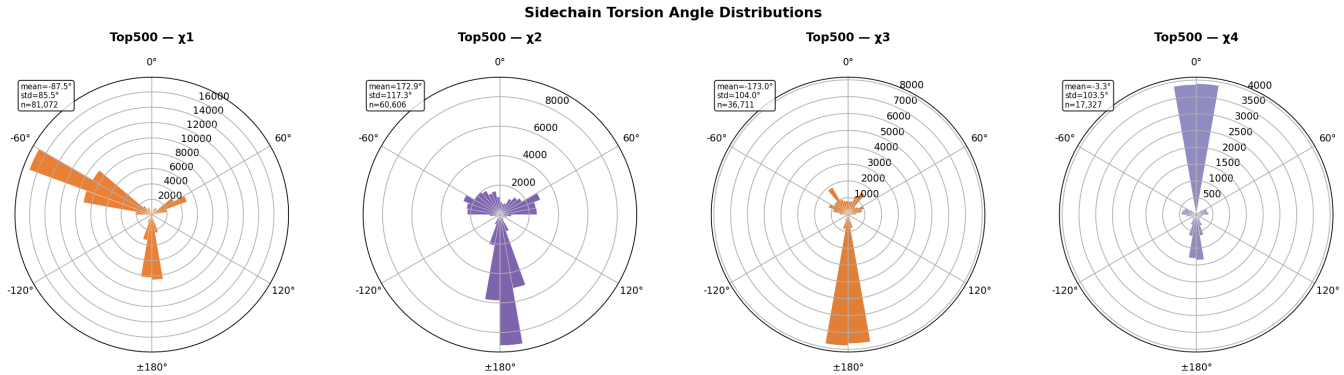
3.1.1 Backbone Torsion Angles (φ, ψ, ω)



3.1.2 Omega Deviation from Planarity (Trans Peptides Only)

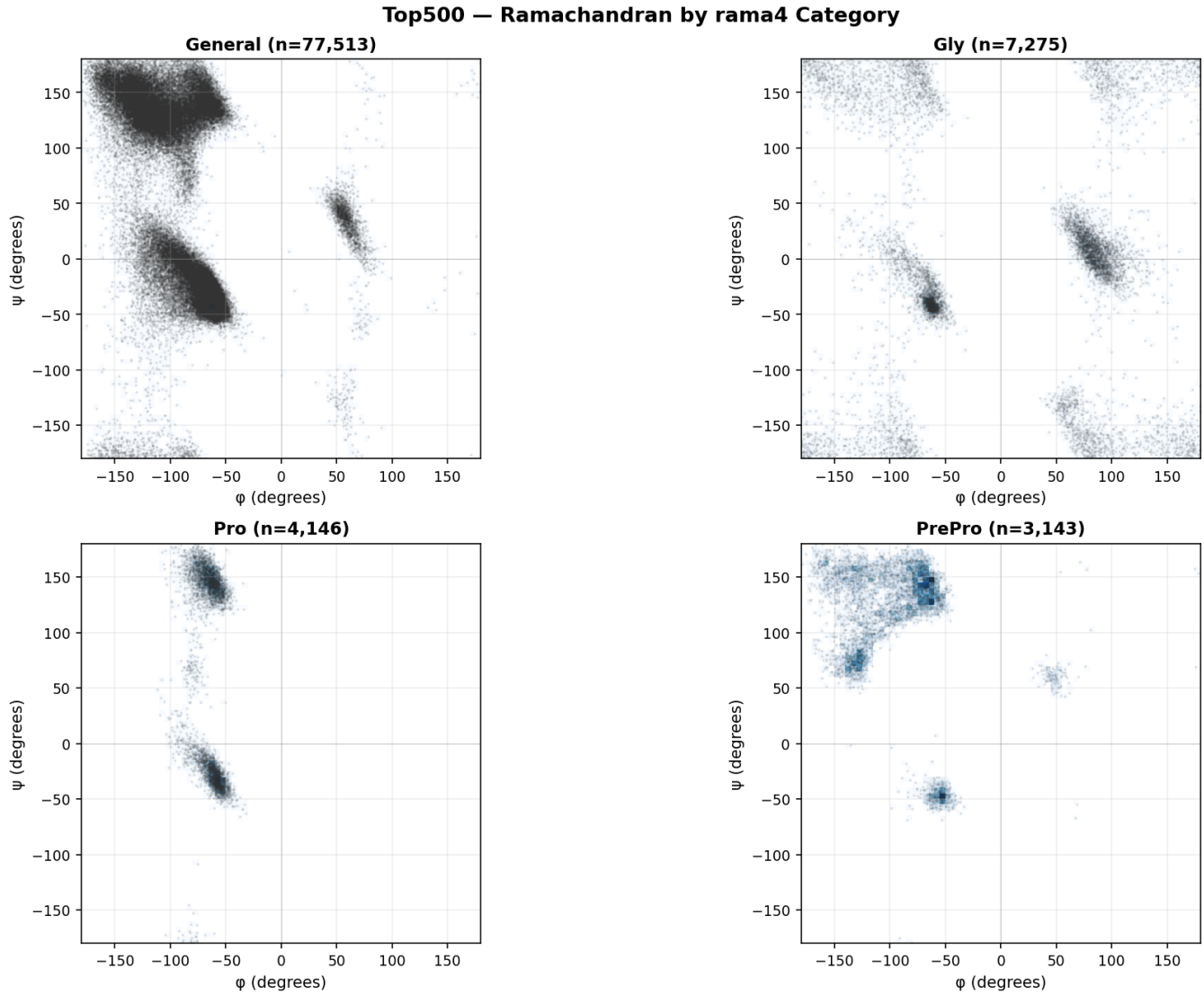


3.1.3 Sidechain Torsion Angles (χ1–χ4)



3.2 Multivariate Distributions

3.2.1 Ramachandran Distributions (φ × ψ by Category)

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Category counts (rama4):
  General        77,513  (79.49%)
  Gly             7,275  ( 7.46%)
  Pro             4,146  ( 4.25%)
  PrePro          3,143  ( 3.22%)
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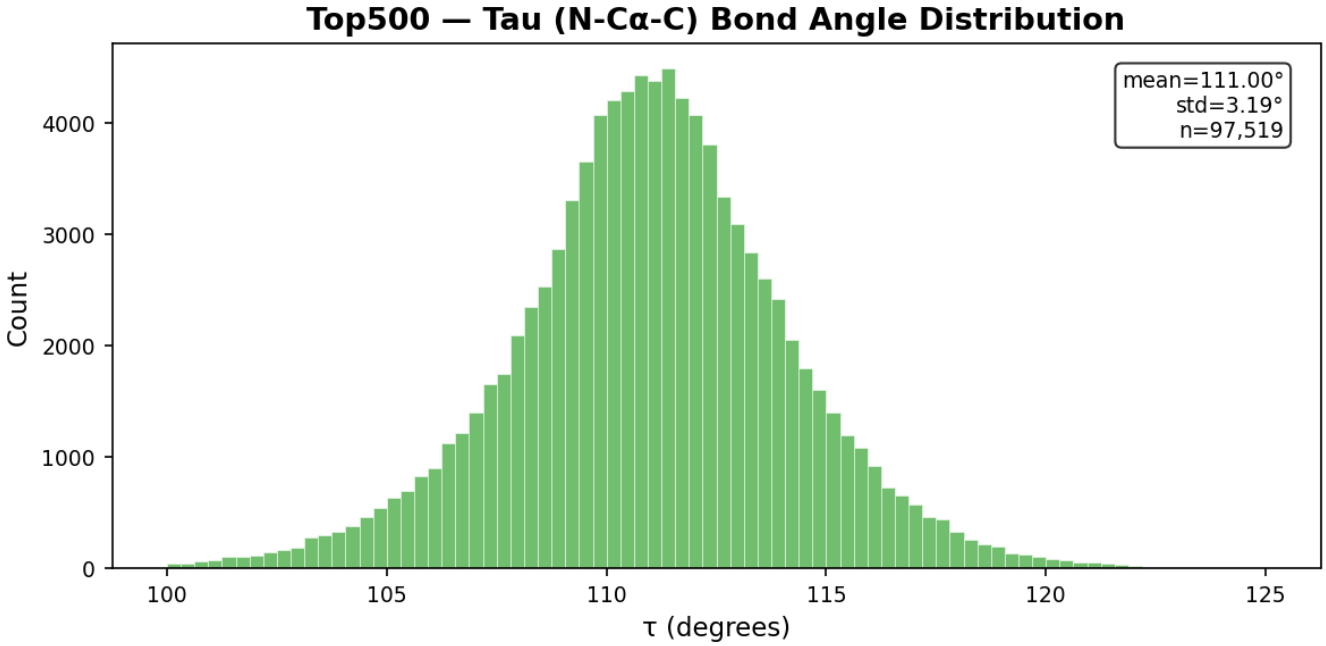
4. Bond Angle Distributions (Linear Statistics)

Bond angles are non-periodic and are analyzed with standard linear statistics (arithmetic mean, standard deviation).

4.1 Tau (N−Cα−C)


Top500 — Tau (N-Cα-C) Bond Angle Distribution
mean=111.00°
std=3.19°
n=97,519
Count
0
1000
2000
3000
4000
100
105
110
115
120
125
τ (degrees)

5. References

- Lovell, S. C., Davis, I. W., Arendall, W. B. III, de Bakker, P. I. W., Word, J. M., Prisant, M. G., Richardson, J. S., & Richardson, D. C. (2003). Structure validation by Cα geometry: φ, ψ and Cβ deviation. *Proteins*, 50(3), 437–450. doi:10.1002/prot.10286
- Lovell, S. C., Word, J. M., Richardson, J. S., & Richardson, D. C. (2000). The penultimate rotamer library. *Proteins*, 40(3), 389–408. doi:10.1002/1097-0134
- Hintze, B. J., Lewis, S. M., Richardson, J. S., & Richardson, D. C. (2016). Molprobity's ultimate rotamer-library distributions. *Proteins*, 84(9), 1177–1189. doi:10.1002/prot.25039
- Williams, C. J., Headd, J. J., Moriarty, N. W., Prisant, M. G., Videau, L. L., Deis, L. N., Verma, V., Keedy, D. A., Hintze, B. J., Chen, V. B., Jain, S., Lewis, S. M., Arendall, W. B. III, Snoeyink, J., Adams, P. D., Lovell, S. C., Richardson, J. S., & Richardson, D. C. (2018). MolProbity: More and better reference data for improved all-atom structure validation. *Protein Science*, 27(1), 293–315. doi:10.1002/pro.3330
- Word, J. M., Lovell, S. C., Richardson, J. S., & Richardson, D. C. (1999). Asparagine and glutamine: using hydrogen atom contacts in the choice of side-chain amide orientation. *Journal of Molecular Biology*, 285(4), 1735–1747.
- Davis, I. W., Leaver-Fay, A., Chen, V. B., Block, J. N., Kapral, G. J., Wang, X., Murray, L. W., Arendall, W. B. III, Snoeyink, J., Richardson, J. S., & Richardson, D. C. (2007). MolProbity: all-atom contacts and structure validation for proteins and nucleic acids. *Nucleic Acids Research*, 35(Web Server), W375–W383. doi:10.1093/nar/gkm216
- Edison, A. S. (2001). Linus Pauling and the planar peptide bond. *Nature Structural Biology*, 8(3), 201–202. doi:10.1038/84921

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